01

LECTURE

HUMAN HISTOLOGY

MT 2, First Semester LECTURE 01 - Dr. Ma. Lourdes Bajarias

CELL STRUCTURE

TOPIC OVERVIEW

LEARNING OBJECTIVES

- Learn the generalities, morphology and functions of the cell
- Enumerate and describe its components of the cytoplasm and the different organelles
- To relate the significance to the overall function of the organism

I. THE CELL

- The cell is the **basic and structural unit** of all multicellular organisms.
- It has several cellular processes:
  - Absorption of metabolites
  - Ingestion
  - Digestion
  - Elimination of metabolic wastes
  - Protection
  - Movement
  - Reproduction
  - Death
- It can be distinguished through:
  - Size
  - Shape
  - Nucleus
  - Cytoplasm
  - Arrangement

A. GENERALITIES OF THE CELL

Shape

- Isolated cell: **spherical**
- Surface contact can flatten a cell
- Factors that can modify the shape of a cell:
  - **The cell has a specialized function**
    - Nerve cell is **stellate** or **angled** due to its cellular processes
      - ➔ **Stellate**: Star shaped
      - ➔ Specialized function: **Axons and Dendrites**
    - Muscle is spindle shaped
      - ➔ **Spindle** shaped: elongated with thick middle part
      - ➔ Specialized function: **Contraction**
  - **The cell is exposed to stressors**
    - Example: The cell is exposed in a hypertonic solution, the cell will shrink.

The Plane of the Section

- The cell to be seen differently based on how the specimen is cut.
- Could be cut across, cross-sectionally, or diagonally.

Size

- It is difficult to measure a cell without the use of special gadgets or instruments.
- Instead, we can compare its size to a cell that is most commonly encountered by MedTechs, the Red Blood Cell.
- The cell is in contact with a surface or neighboring cells
- Epithelial cells flatten when in contact with a surface and neighboring cells.

Polarity

- This is the cell's definite spatial arrangement of some parts of the cell.
- Example: Cell polarity refers to the presence of distinct basal and apical surfaces. The **basal surface** rests on the basement membrane, while the **apical surface** faces the free space. The nucleus is typically located near the basal surface, reflecting the cell's polarity.

Orientation

- This points to the axis of the cell
- This refers to the plane of section, or how the tissue was cut before fixation and mounting on the slide.

Specialization

- Cells develop different structures correlated to a specific function.

B. CELL STAINING

In histology, the stains (routine histologic stains) used are:

Eosin

- It stains the basic cellular components, the cellular components stained are often called **acidophilic** and are seen as **pink to red** in the microscope.
    - Basic cytoplasmic components
    - Filaments
    - Cytoskeleton
    - Collagen fibers
    - RBC

Hematoxylin

- It stains the acidic cellular components, the cellular components stained are often called **basophilic** and are seen as **blue to purple** in the microscope.
    - Nucleus
    - DNA
    - RNA
    - Heterochromatin
    - Nucleoli
    - Acidic cytoplasmic components
    - Cartilage matrix

EXTRA NOTE!
Based on Doc, Ma. Lourdes Bajarias, Hematoxylin is not considered a basic dye or basic stain, same with eosin which isn't considered an acidic dye. They only BIND TO ACIDIC (Hematoxylin) OR BASIC (Eosin) cellular components

C. COMPARTMENTS OF THE CELL

The cell is composed of two compartments: the nucleus and the cytoplasm.

- **Nucleus:** is the largest organelle within the cell and contains the genome along with the enzymes necessary for DNA replication and RNA transcription.
    - Chromatin, Nucleolus, Nucleoplasm, Nuclear membrane
- **Cytoplasm:** is the part of the cell located outside the nucleus. The cytoplasm contains organelles("little organs"), cytoskeleton (made of polymerized proteins that form microtubules, intermediate filaments, and actin filaments), and inclusions suspended in an aqueous gel called the **cytoplasmic matrix.**
    - Organelles (membranous and nonmembranous), Cytoskeleton, and Inclusions

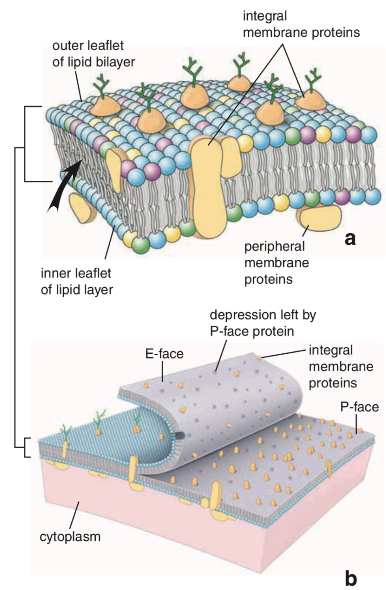
CYTOPLASM				NUCLEUS
Organelles		**Cysto skeleton**	**Inclusions**	Chromatin, Nucleolus,
Membranous	**Non-membranous**			Nucleoplasm, Nuclear Membrane
Plasma membrane, Endoplasmic reticulum, Golgi apparatus, Mitochondria, Endosomes, Exosomes, Lysosomes, Peroxisome, Transport Vesicles	Microtubules, Filaments, Centrioles, Ribosomes, Proteasomes			

Table 1. Cell Compartments

II. CYTOPLASM: MEMBRANOUS ORGANELLES

A. PLASMA MEMBRANE

- It is also known as **Cell Membrane** or **Plasmalemma**



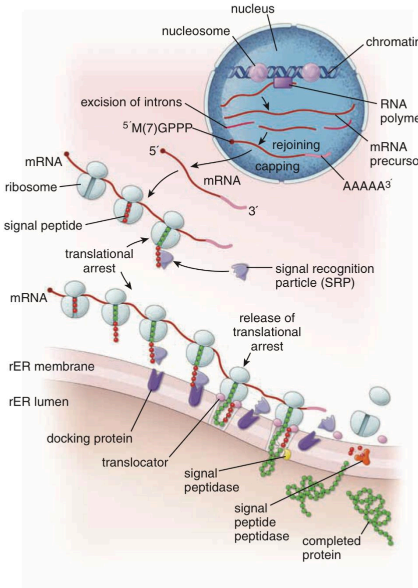
- The plasma membrane is composed of:

- **Amphipathic phospholipid bilayer**
  - Phospholipid molecules have polar, hydrophilic (water-loving) heads and nonpolar, hydrophobic fatty-acid tails.
  - The hydrophilic heads face the aqueous environments on both sides of the membrane, forming the outer and inner leaflets, while the hydrophobic tails face inward toward each other, creating the membrane's core.
  - The interior of the bilayer is hydrophobic, water and other polar molecules cannot easily pass through the middle of the membrane. Therefore, they require helper proteins, such as channels or carriers, to cross the membrane.
- **Integral and Peripheral Proteins**
  - **Integral proteins** are embedded within the layer
  - **Peripheral proteins** are not embedded within the layer but are connected through a strong ionic force
  - **Membrane proteins** function as channels, pumps, receptors, enzymes, and structural components of the plasma membrane.
- **Cholesterol**
  - Incorporated within the gaps between phospholipids

- These molecules are arranged in a mosaic pattern, which is why the membrane is described by the **Modified fluid mosaic model.** This model states that these components form a mosaic embedded in a fluid, lipid bilayer.
- **Glycocalyx (Cell Coat)**
  - This is the outer layer of the membrane
  - The glycocalyx is composed of glycoproteins (carbs + protein) and glycolipids (carbs + lipid) that contain carbohydrate chains.
  - It serves as a cellular "ID" for cell recognition, plays a role in cell interactions and metabolism, and functions as receptor sites for signaling molecules.
- **Lipid Rafts**
  - They are small, thicker microdomains within the plasma membrane that act as platforms for cellular processes.
  - They are enriched with **cholesterol and specific lipids**, which makes them more ordered and less fluid than the surrounding membrane.
  - These lipid rafts make the plasma membrane **not completely fluid**, but instead have regions with different degrees of fluidity.
- A plasma membrane is not visible with a light microscope (but may be represented by **a** line) but visible with transmission electron microscope. In an electron microscope, it is seen as two inner and outer electron-dense layers separated by electron-lucent layers. Some cells may have extensive infoldings.

B. ROUGH ENDOPLASMIC RETICULUM

- Series of interconnected, membrane-limited, flattened sacs called **cisternae** with attached *ribosomes* on the surface (that's why it's rough)
  - *Ribosomes* are non-membranous organelles and contain ribosomal RNA and various proteins
  - Groups of ribosomes form polyribosomes/polysomes
  - Where the translation and post-translation steps of protein synthesis occur
- Usually continuous with the nucleus
- Represented by the **ergastoplasm,** the area of the cell that stains intensely with basic dye
  - So if you see a deep deep basophilic area near the nucleus, that might be the RER
- Serves as quality checkpoint and is highly developed in active secretory cells
  - Cells that keeps on producing proteins and secretions are those who have very developed rough ER

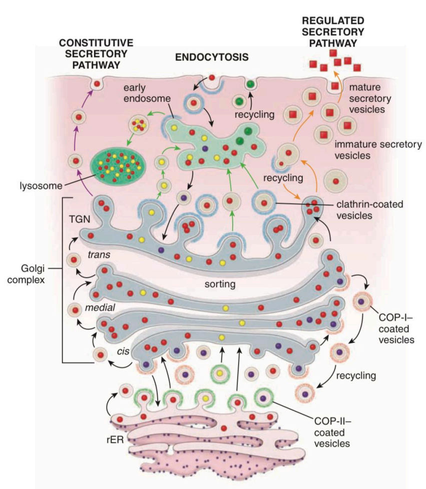


C. SMOOTH ENDOPLASMIC RETICULUM

- Made of **anastomosing short tubules** with **no ribosomes**
- Represented by cytoplasmic acidophilia in high amounts
- May or may not be separate from the rER
- **Functions:**
  - Glycogen Metabolism
  - Detoxification of xenobiotics (or any substances taken in that came from outside of the body e.g. medicine or drugs)
  - Membrane formation and recycling
- Called the **sarcoplasmic reticulum** in skeletal and cardiac muscles
  - Skeletal and cardiac muscles are **acidophilic** because they are rich in smooth ER

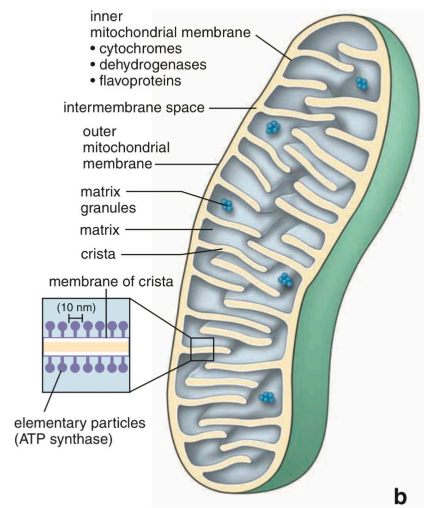
D. GOLGI APPARATUS

- Series of **stacked**, flattened, membrane-limited sacs or "cisternae" and tubular extensions found near the Microtubule Organizing Center (MTOC)
  - **cis-Golgi network:** near rER representing the forming face
  - **trans-Golgi network:** away from rER representing the maturing face
  - **medial-Golgi network:** in between
- Proteins are packed and sorted in the Golgi Apparatus and each area differs in maturity.
- It slowly matures and is then sent out of the cytoplasm and to the area where the protein is needed.
- We can think of it like the "*Shenzhen Sorting Center*"
- **Function:** post-translational modification, sorting, and packaging of proteins
- Light microscope: **clear area** partially surrounded by ergastoplasm in cells with large GA
- Active in **secretory cells** and cells that synthesize large amounts of membrane and **membrane-associated proteins**



E. MITOCHONDRIA

- Contain the enzyme system that generates ATP for energy
- No definite shape and number within a cell but has constant characteristic of **membranes**
- 2 membranes with a space in between called intermembranous space
- Possess their own genome (mitochondrial DNA) and has its own cellular processes
  - Mitochondrial DNA comes from mothers
  - DNA from nucleus is from both parents
- Can replicate and make it's own protein
- It filters what comes inside and has different enzymes per layer



Outer mitochondrial membrane

- Porins/Voltage-dependent anion channels
- Permeable to uncharged molecules ≤ 5 kDa
- Allows small molecules, ions and metabolites to pass through
- Contains phospholipase A2, monoamine oxidase, and acetyl-CoA synthase

Inner mitochondrial membrane

- Thinner and arranged into folds **(cristae)** to increase surface area
- Rich in the phospholipid **cardiolipin** making the membrane *impermeable to ions*
- Contains the respiratory transport chain enzymes which produce a lot of atp

Intermembrane space

- Located between the inner and outer membranes
- Contain creatine kinase, adenylate kinase, and cytochrome c

Matrix

- **Middle of the Mitochondria**
- Contains the enzymes for the Krebs cycle, fatty acid f-oxidation, mtDNA, ribosomes and tRNAs
- Mitochondria secures the matrix to avoid the disruption of metabolic processes happening within

- Present in all cells **except RBCs and terminal keratinocytes or skin cells** since they no longer have organelles
  - RBC gets energy from Glycolysis
  - Terminal Keratinocytes are dead cells and has no need for energy production = no mitochondria
- More numerous in cells that use **large amounts of energy** such as muscles and neurons
- Contributes to (Cytoplasmic) **acidophilia** when present in large amounts

- May be stained by **histochemical procedures** that demonstrate its enzymes

F. ENDOSOMES

- Network of **membrane-enclosed cytoplasmic compartments** that sort endocytosed material and membrane proteins
- Substances that enter our cells and enclosed by endosomes
  - **Endocytosis**: process of vesicular transport when substances enter the cell
- Start as **early endosomes**/early-sorting endosomes →
  late endosomes/late-sorting endosomes → multivesicular bodies/ **multivesicular endosomes** →
  either become lysosomes OR exosomes
  - Endosomes can also bind to lysosomes to degrade its contents through the **HYDROLYTIC** enzymes of lysosomes

Early Endosomes

- **Near the cell membrane**
- Tubulovesicular structure
- Proteins are sorted into (a) those transported to late endosomes and (b) those for recycling
- **Slightly more acidic** environment than the cytoplasm pH 6.2-6.5

Late Endosomes

- Near the nucleus and the Golgi apparatus
- More complex structure with **onion-like internal membranes**
- Will form **multivesicular bodies** by inward invaginations
- **More acidic** (pH 5.5)

G. EXOSOME

- Very small, endosome-derived vesicles **released by cells** into the extracellular environment
- Formed when MVB's (Multivesicular Bodies - Endosome) fuse with the plasma membrane
- Present in high concentrations in all **body fluids**
- Contain nucleic acids, proteins, lipids, cytokines, transcription factor receptors, growth factors, and other bioactive substances

H. LYSOSOME

- Rich in **hydrolytic enzymes** that degrade macromolecules derived from endocytosis
- Digest parts of the cell itself, process called **autophagy**
- The membrane of the lysosome does not get degraded by its own hydrolytic enzymes due to two reasons:
  - Presence of lysobisphosphatidic acid
  - High glycosylation (Glycosylated Membrane Proteins) of the luminal surface
- Has proton (H^{*}) pumps that regulates the acidity of lysosomes which is a very important factor in its hydrolytic enzyme degradation function
- Recognizable in the light microscope in some cells because of their number, size, or contents
- Residual bodies: debris-filled vacuoles

I. PEROXISOME

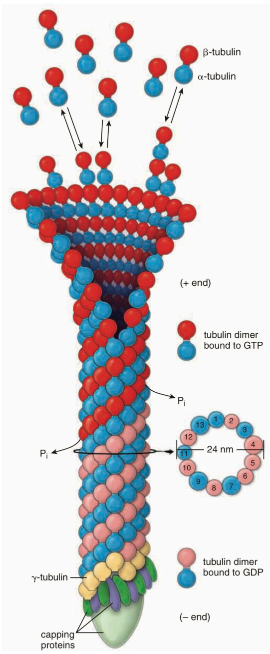
- Small and spherical, distributed throughout the cytoplasm often with electron-dense crystalloid inclusions
- Contains oxidative enzymes found in almost all cells in the body and increased in response to diets, drugs, and hormonal stimulation.
- Found in almost all cells of the body; abundant in liver and kidney; cell increases in response to diet, drugs, and hormonal stimulation
- Functions: Oxidative digestion, lipid biosynthesis, metabolism of reactive oxygen intermediates
- Visible only after special histochemical stains.

III. CYTOPLASM: NONMEMBRANOUS ORGANELLES

- Mostly part of the cytoskeleton (structure of the cells).

A. MICROTUBULES

- Microtubules are **nonbranching rigid hollow tubes of polymerized proteins** that can assemble or disassemble.
- Cylindrical in form, originates from the microtubule-organizing center (MTOC) with a diameter of 20-25 nm.
- Found near the nucleus and extends to the periphery.
- Components of several specialized cell structures, including:
  - Cillia
  - Flagella
  - Centrioles
  - Mitotic spindles
  - Elongating processes (pseudopods)
- Used for transport, they extend or shorten which serves as movement for the organelles.



- Some used for movement of the cells themselves and the attachment of ribosomes to the mitotic spindles
- Maintains cell shape.
- Guides endosomes and lysosomes towards different areas of cells.
- Have a non-growing end (negative end) which is found near the MTOC and have a growing (positive end) which extends to the periphery. **They can only grow on one end.**
- Noticed by using special stains or immunocytochemicals.

B. ACTIN FILAMENTS

- Protein molecules that polymerize spontaneously into a linear helical array to form filaments with diameter of 6-8 nm.
- F-actin (filamentous actin): Polymerized chains of actin.
- G-actin (globular actin): Free, unpolymerized actin.
- Thinner, shorter, and more flexible than microtubules
- One of its functions involves locomotion at the muscle tissue level.
- Polarization of actin filaments:
  - Plus (+) end = Fast-growing end
  - Minus (-) end = Slow-growing end

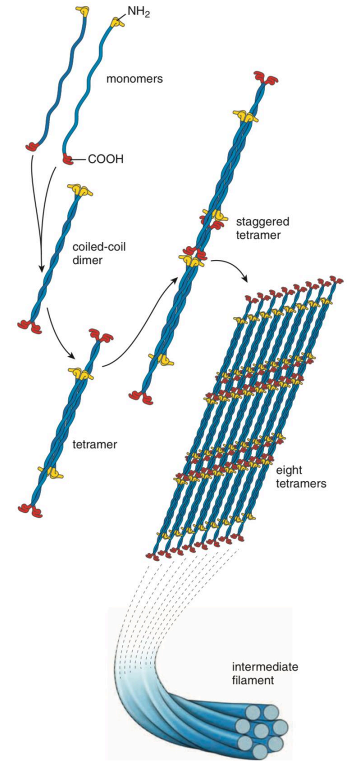
  Actin filaments can grow on both ends
- **Mechanism: Treadmilling**
  - Actin is added to the plus (+) end of the filament, and dissociates from the minus (-) end resulting in movement.

C. INTERMEDIATE FILAMENTS

- Ropelike, nonpolar, with no enzymatic activity.
- Does not assemble nor disassemble; structure is maintained.
- Nonpolar, no plus or minus end.
- Function: To hold the structure of the cell.
- Monomers are twisted together to form a coiled dimer which together are the intermediate filaments.

Six classes of Intermediate Filaments:

1. Acid cytokeratin
2. Basic cytokeratin
3. Vimentin and vimentin-like
4. Neurofilaments
5. Lamins
6. Beaded filaments

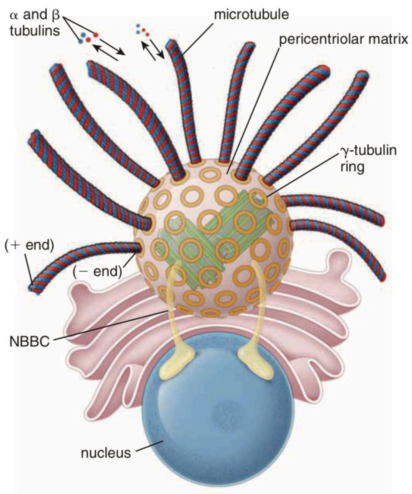


D. CENTRIOLES

- Paired, short, rod-like cytoplasmic cylinders built from microtubules.
- Where the microtubule organizing center assembles (MTOC).
- Orientation is called orthogonal, forming 90 degrees with each other.
- We can find centrioles close to the nucleus surrounded by a zone of dense amorphous pericentriolar material which serves as their connection.
- Visible under a light microscope, represented as two dots.

E. MICROTUBULE-ORGANIZING CENTER (MTOC)

- Controls the number, polarity, direction, and orientation of microtubules.
- **Orthogonal orientation** at rest.



F. BASAL BODIES

- Important for the development of the cilia (hair-like cell structures) which requires the presence of basal bodies.
- Derived from the centrioles.

G. RIBOSOMES

- Important for protein synthesis.

H. PROTEASOMES

- Protease complexes.
- Enzymatically degrade damaged and unnecessary proteins into small polypeptides and amino acids.

I. INCLUSIONS

Lipofuscin

- "Wear and Tear" pigment
- Accumulate as the cell grows old
- **Brownish-gold** pigment in H&E
- Made of oxidized lipids, metals, and organic molecules
- Easily seen in nondividing cells
- Also seen in phagocytic cells
- May be an accurate indicator of cellular stress

Hemosiderin

- **Iron-storage complex** in the cytoplasm
- Indigestible residues of hemoglobin
- Seen as **deep brown** granules but iron can be special stained via histochemical methods
  - often mistaken for lipofuscin due to its similar color
- Easily seen in the spleen

Glycogen

- Storage material for glucose
- Not visible in the light microscope unless with special fixatives and stains
  - eg. toluidine blue or the periodic acid -Schiff [PAS] method
- Unstained regions in large amount like in liver and striated muscle
- In electron microscope, appear as clusters of granules in the cytoplasm

Lipid Inclusions (Fat droplets)

- Provide energy for cellular metabolism
- **Adipocytes:** Fat cells
- Seen as a space or hole which represents the space where the lipid once was
- Extracted by organic solvents used in preparing tissues
- Seen as a space or hole which represents the space where the lipid once was

Crystalline Inclusions

- Specific for certain cells under the light microscope
  - Example:
  - Sertoli & Leydig cells of the testis
- May be found in any part of the cell
- May contain viral proteins, storage material, or cellular metabolites, others not clear

J. CYTOPLASMIC MATRIX

- Also known as ground substance or cytosol
- It is a concentrated aqueous gel containing molecules of different sizes and shapes
- Site of the cell's physiologic processes
- Under a HIGH-VOLTAGE ELECTRON MICROSCOPE (HVEM)**:** appears as a complex three-dimensional structural network of thin microtrabecular strands and crosslinkers

IV. NUCLEUS

- Control center, directs all functions and metabolic processes of the cell.
- Contains genetic information and machinery for replication and transcription.
- Reads code, replicates dna, transcribes mRNA for proteins
- Has a nucleolus in the center, enveloped in a nuclear envelope

Components of the Nucleus
**In a nondividing cell or at interphase*

- **Chromatin** - DNA and proteins
- **Nucleolus** - Site of rRNA synthesis
- **Nuclear Envelope** - Double membrane system
- **Nucleoplasm** - The rest of the nuclear components

A. CHROMATIN

- Provides energy for cellular metabolism

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<tr><th colspan="2">TYPES OF CHROMATIN</th></tr>
<tr><th>HETEROCHROMATIN</th><th>EUCHROMATIN</th></tr>
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Highly condensed chromatin
Densely stained clumps
Types:

Constitutive - Repetitive DNA sequence
Facultative - Non-repetitive sequence

Stains with hematoxylin and basic dyes
Prominent in metabolically inactive cells

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Lightly staining material where most genes are located

stains lightly because DNA is “uncoiled” and ready for transcription

Not evident in the light microscope
Indicates active chromatin
Prominent in metabolically active cells
Vesicular nucleus

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B. NUCLEOLUS

- The nonmembranous region of the nucleus
- Site of ribosomal RNA (rRNA) synthesis and ribosome assembly
- Stains intensely with hematoxylin and basic dyes and metachromatically with thionine dyes
- Well-developed in cells active in protein synthesis
- Involved in cell cycle regulation
- Varies in size and number

C. NUCLEAR ENVELOPE

- Formed by two membranes with a perinuclear cisternal space in between
- Provides a selectively permeable barrier between the nuclear compartment and the cytoplasm
- Outer membrane resembles and is continuous with the membrane of the endoplasmic reticulum
- Inner membrane is supported by a rigid network of intermediate protein filaments called the nuclear lamina
- Membranes are perforated at intervals by nuclear pores

D. NUCLEOPLASM

- Material enclosed by the nuclear envelope aside from the chromatin and the nucleolus
- Described to be amorphous
- May contain crystalline, viral, and other inclusions
  - **REVIEW**: Cell Division, Phases of the Cell Cycle

Nuclear number, shape & size

- Commonly, there is only one nucleus in a cell; But some can have 2 (binucleate) such as some cells in the liver, bladder, cardiac muscle; While some have several nuclei (multinucleate) such as in skeletal muscle fibers, giant cells of developing bone and in some ganglion
- Nucleus is usually globular to ovoid, in conformity with the cell shape; cytoplasmic inclusions may distort the nuclear shape; some have special shapes (eg. elongate, crescentic, lobate)
- Nuclear size is relative to the cell size and is uniform in the same cell type

Cell Death

- Result from accidental cell injury or mechanisms that cause cells to self-destruct
- Mechanisms of cell death based on signaling mechanism:
  - Nonprogrammed cell death/Necrosis
  - Programmed cell death
    - Apoptotic cell death
    - Nonapoptotic cell death

Features of Dying Cells	Necrosis	Apoptosis
Cell Swelling	+	-
Cell Shrinkage	-	+
Damage to the Plasma Membrane	+	-
Plasma Membrane Blebbing	-	+
Aggregation of Chromatin	-	+
Fragmentation of the Nucleus	-	+
Oligonucleosomal DNA Fragmentation	-	+
Random DNA Degradation	+/-	-
Release of Cytochrome C and SMAC/DIABLO from	-	+

Mitochondria		
Caspase Cascade Activation	-	+

Dr. Bajarias mentioned that she will focus on the section highlighted in red

Nuclear alterations in dying cells

- **Karyolysis** - disappearance of nucleus due to the dissolution of the DNA by increased DNAse activity
- **Pyknosis** - shrinkage of the nucleus due to chromatin condensation
- **Karyorrhexis** - fragmentation of the nucleus

REFERENCES

- Pawlina, W. (2024). Histology: A text and atlas: With correlated cell and molecular biology (9th ed.). Philadelphia: Wolters Kluwer Health
- Arey, L.B. (1974). Human histology (4th ed.). Philadelphia: W.B. Saunders Company
- Histology Laboratory Manual

(Histology: A Text and Atlas 7th Edition, Pawlina)

Classes of Intermediate Filaments (p. 76)

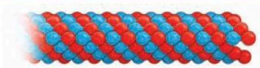
TABLE 2.3 Classes of Intermediate Filaments, Their Location, and Associated Diseases

Type of Protein	Molecular Weight (kDa)	Where Found	Examples of Associated Diseases
Class 1 and 2: Keratins			
Acid cytokeratins	40–64	All epithelial cells	Epidermolysis bullosa simplex
Basic cytokeratins	52–68	All epithelial cells	Keratoderma disorders caused by keratin mutations Meesman corneal dystrophy
Class 3: Vimentin and Vimentin-Like			
Vimentin	55	Cells of mesenchymal origin (including endothelial cells, myofibroblasts, some smooth muscle cells) and some cells of neuroectodermal origin	Desmin-related myopathy (DRM) Dilated cardiomyopathy Alexander disease Amyotrophic lateral sclerosis (ALS)
Desmin	53	Muscle cells; co-assembles with nestin, synemin, and paranemin	
Glial fibrillary acidic protein (GFAP)	50–52	Neuroglial cells (mainly astrocytes; to lesser degree, ependymal cells), Schwann cells, enteric glial cells, satellite cells of sensory ganglia, and pituicytes	
Peripherin	54	Peripheral neurons	
Class 4: Neurofilaments			
Neurofilament L (NF-L)	68	Neurons Co-assembles with NF-M or NF-H	Charcot-Marie-Tooth disease Parkinson disease
Neurofilament M (NF-M)	110	Neurons Co-assembles with NF-L	
Neurofilament H (NF-H)	130	Neurons Co-assembles with NF-L	
Nestin	240	Neural stem cells, some cells of neuroectodermal origin, muscle cells Co-assembles with desmin	
α-Internexin	68	Neurons	
Synemin α/β[a]	182	Muscle cells Co-assembles with desmin	
Syncoilin	64	Muscle cells	
Paranemin	178	Muscle cells Co-assembles with desmin	
Class 5: Lamins			
Lamin A/C[b]	62–72	Nucleus of all nucleated cells	Emery-Dreyfuss muscular dystrophy
Lamin B	65–68	Nucleus of all nucleated cells	Limb girdle muscular dystrophy
Class 6: Beaded Filaments			
Phakinin (CP49)[c]	49	Eye lens fiber cells Co-assembles with filesin	Juvenile-onset cataracts Congenital cataracts
Filesin (CP115)	115	Eye lens fiber cells Co-assembles with phakinin	

[a] Synemin α and synemin β represent two alternative transcripts of the DMN gene.
[b] Lamin C is a splice product of lamin A.
[c] The molecular weight of filensin/phakinin heterodimer is 131 kDa.

Summary Characteristics of the Three Types of Cytoskeletal Elements (p. 83)

TABLE 2.4 Summary Characteristics of Three Types of Cytoskeletal Elements

	Actin Filaments (Microfilaments)	Intermediate Filaments	Microtubules
			
Shape	Double-stranded linear helical array	Rope-like fibers	Nonbranching long, hollow cylinders
Diameter (nm)	6–8	8–10	20–25
Basic protein subunit	Monomer of G-actin (MW 42 kDa)	Various intermediate filament proteins (MW ~50 kDa)	Dimers of α- and β-tubulin (MW 54 kDa); γ-tubulin found in MTOC is necessary for nucleation of microtubules; δ-, ϵ-, ζ-, η-tubulins are associated with MTOC and basal bodies
Enzymatic activity	ATP hydrolytic activity	None	GTP hydrolytic activity
Polarity	Yes; minus (−) or pointed end is slow-growing end Plus (+) or barbed end is faster-growing end	Nonpolar structures	Yes; minus (−) end is nongrowing end embedded in MTOC Plus (+) end is the growing end
Assembly process	Monomers of G-actin are added to growing filament Polymerization requires presence of K^+, Mg^{2+}, and ATP, which is hydrolyzed to ADP after each G-actin molecule is incorporated into the filament	Two pairs of monomers form two coiled-coil dimers; then two coiled-coil dimers twist around each other to generate a staggered tetramer, which aligns along the axis of the filament and binds to the free end of the elongating structure	At the nucleation site, α- and β-tubulin dimers are added to γ-tubulin ring Each tubulin dimer binds to GTP before it becomes incorporated into the microtubule in the presence of Mg^{2+} After polymerization, GTP is hydrolyzed to GDP
Source of energy required for assembly	ATP	N/A	GTP
Characteristics	Thin, flexible filaments	Strong, stable structures	Exhibit dynamic instability
Associated proteins	Variety of ABPs with different functions: fascin = bundling; gelsolin = filament severing; CP protein = capping; spectrin = cross-linking; myosin I and II = motor functions	Intermediate filament–associated proteins: plectins bind microtubules, actin, and intermediate filaments; desmoplakins and plakoglobins attach intermediate filaments to desmosomes and hemidesmosomes	Microtubule-associated proteins: MAP-1, -2, -3, and -4; MAP-τ; and TOG-ρ regulate assembly, stabilize, and anchor microtubules to specific organelles; motor proteins—dyneins and kinesins—required for organelle movement
Location in cell	Core of microvilli Terminal web Concentrated beneath plasma membrane Contractile elements of muscles Contractile ring in dividing cells	Extend across cytoplasm connecting desmosomes and hemidesmosomes In nucleus just beneath inner nuclear membrane	Core of cilia Emerge from MTOC and spread into periphery of cell Mitotic spindle Centrosome
Major functions	Provide essential components (sarcomeres for muscle cells)	Provide mechanical strength and resistance to shearing forces	Provide network ("railroad tracks") for movement of organelles within cell Provide movement for cilia and chromosomes during cell division

ABP, actin-binding protein; *ADP*, adenosine diphosphate; *ATP*, adenosine triphosphate; *GDP*, guanosine diphosphate; *GTP*, guanosine triphosphate; *kDa*, kilodalton; *MAP*, microtubule-associated protein; *MTOC*, microtubule-organizing center; *MW*, molecular weight; *N/A*, not applicable.